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Design and Software Simulation of a 4-DOF Robotic Arm and Grabber Using ROS2

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Introduction & Motivation

Problem

- Robotic manipulators → nonlinear & uncertain systems
- Affected by: Payload variation, Friction, External disturbances

Limitations of PID

- Fixed gains
- Poor performance under uncertainty
- High tracking error & oscillations

Objective

- Develop adaptive controller (MRAC)
- Achieve robust and accurate tracking

Approach

- MoveIt 2 (Planning) + MRAC (Control)

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System Overview

Hardware & Simulation Stack

A 4-DOF manipulator modeled with full URDF/Xacro specifications — including inertia tensors, mass distributions, and joint limits — simulated in Gazebo under ROS 2 Humble/Jazzy.

Waist

Base rotation (J1)

Shoulder

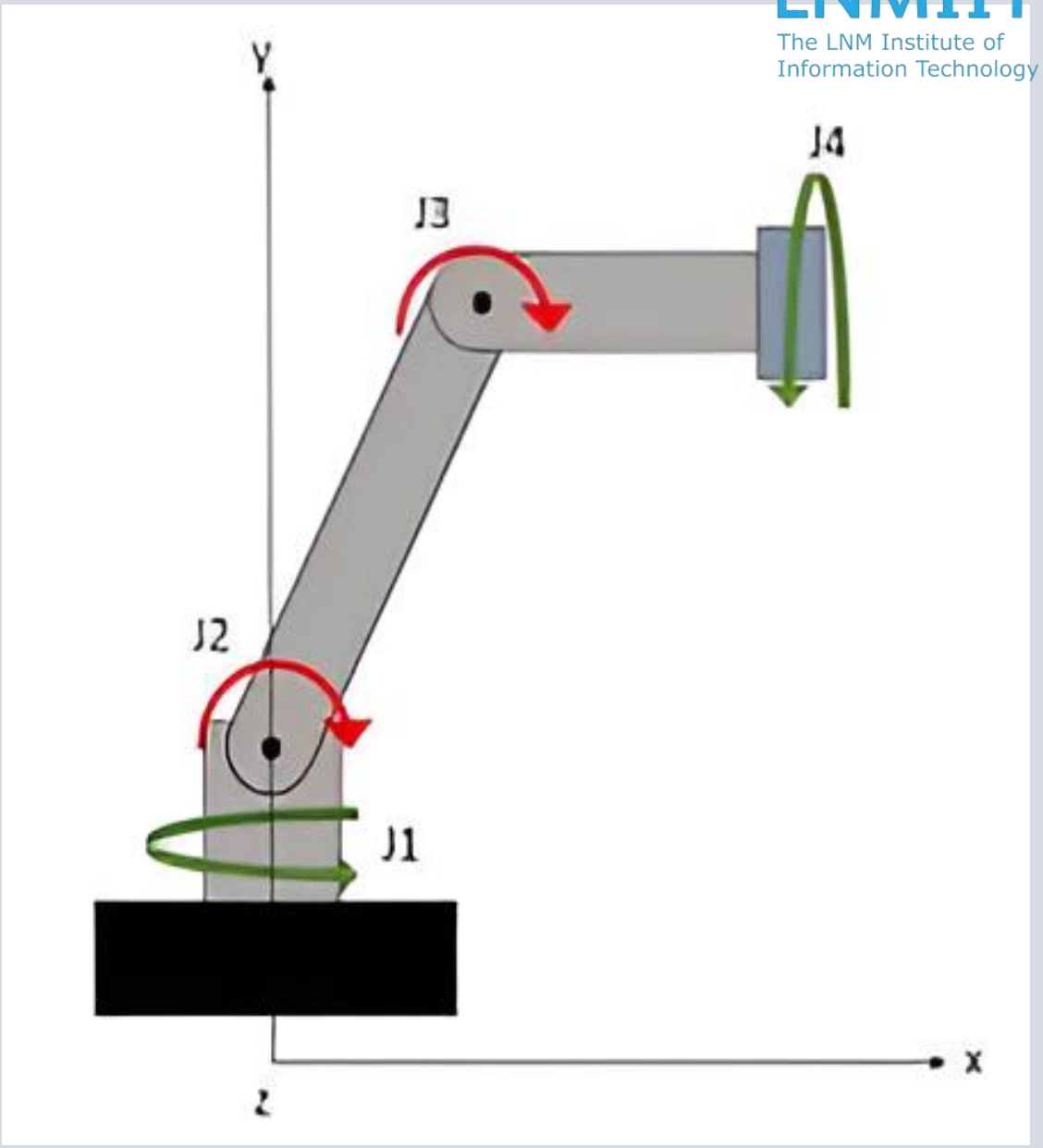
Proximal lift (J2)

Elbow

Mid-link flex (J3)

Wrist

Pitch control (J4)



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Why Model Reference Adaptive Control?

Problem with Classical Control

Fixed-gain PID controllers fail when payload changes, friction varies, or gravitational loading shifts. Performance degrades under uncertainty.

Adaptive Solution

MRAC learns system dynamics online, adjusting control gains K_x and K_r to match a reference model $G_{ref}(s)$.

Robustness Extension

Neural-network-like basis functions ($sin(q)$, $cos(q)$, $tanh(\dot{q})$) compensate for unmodeled friction and hidden dynamics.

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Direct MRAC Control Architecture

Reference Model

Desired second-order dynamics:

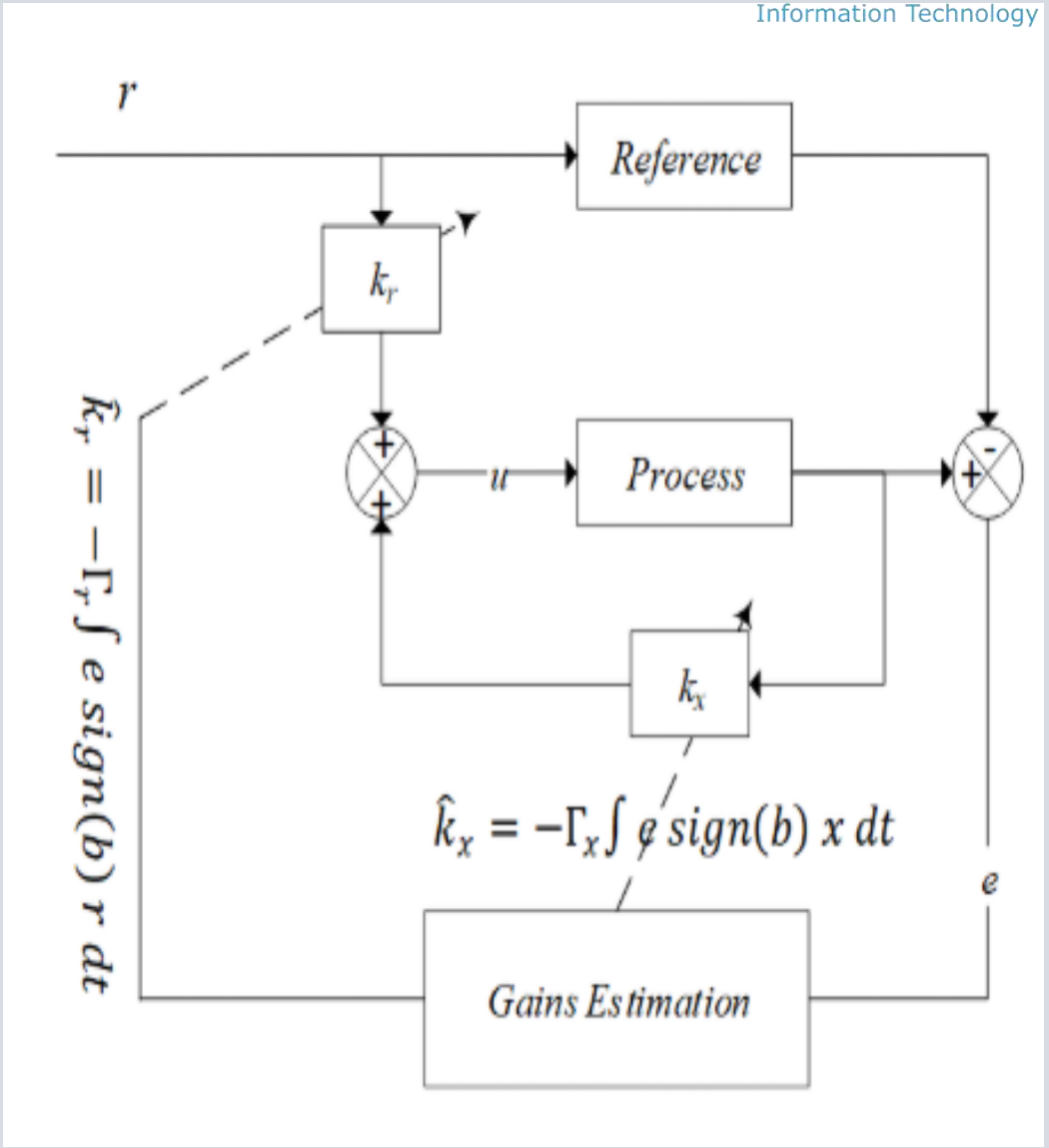
$$G_{ref}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Control Law

Adaptive feedback with learned gains:

$$u = \hat{K}_x q + \hat{K}_r r$$

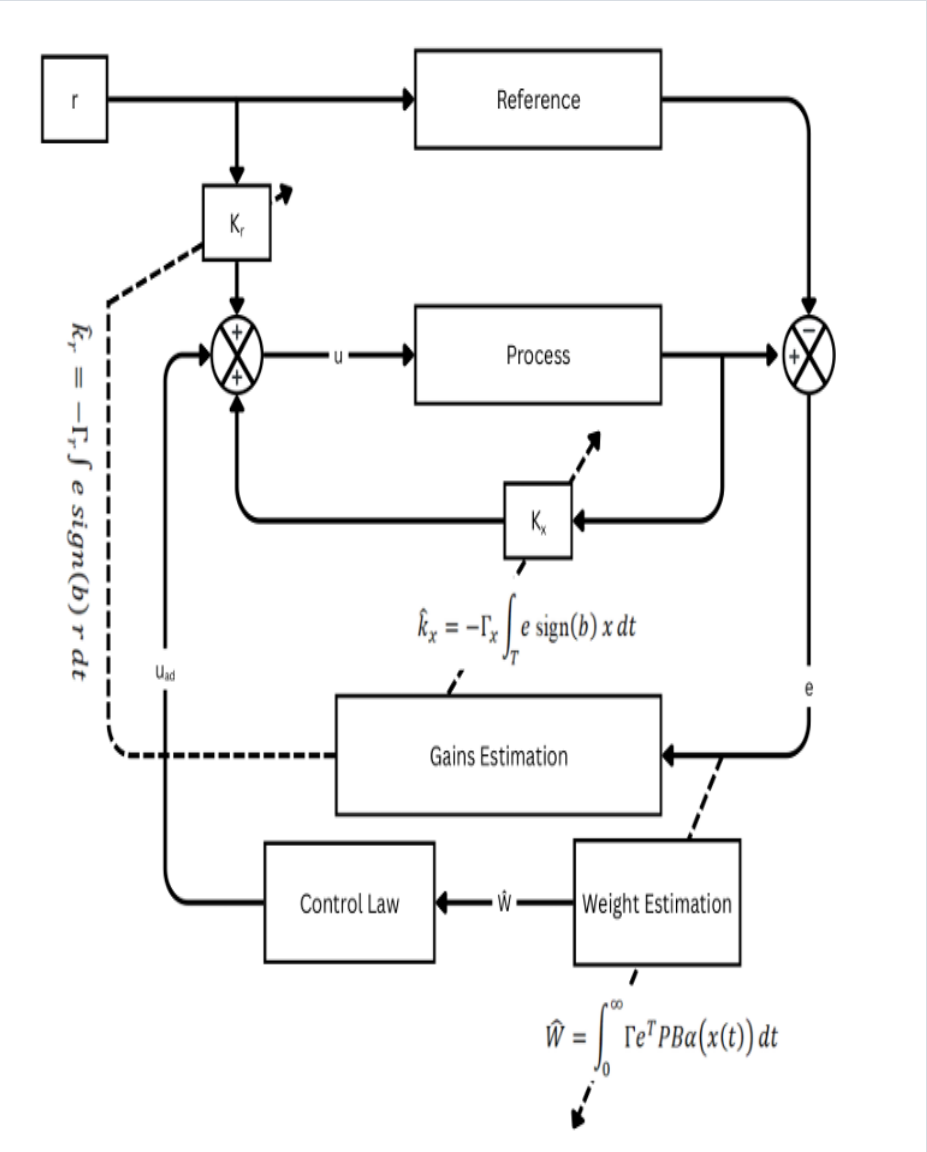
Adaptation follows Lyapunov stability criteria via Barbalat's Lemma.



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Robustness: Handling Unmodeled Dynamics

Robust MRAC augments the baseline controller with a learned compensation term *u_{ad}*, counteracting friction, flexibility, and hidden dynamics.



Augmented Torque Law

$$\tau = u_{mrac} \pm \hat{W}^T \alpha(x)$$

The weight matrix *hat{W}* is updated online. The 22-term basis vector *alpha(x)* spans the likely disturbance space.

Basis Function Terms

sin(q),cos(q)

Gravity & geometric nonlinearities

tanh(dot{q})

Smooth Coulomb friction approximation

Cross-products

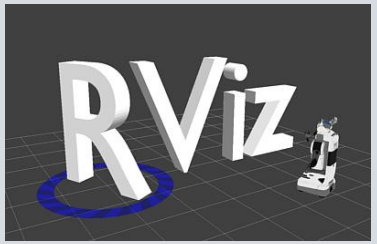
Coriolis & coupling interaction terms

Simulation Setup

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URDF : Unified Robot Description Format



RVIZ : Robot Visualization

ROS2



GAZEBO : A Simulating Tool

MoveIt2

MOVEIT2 : A Motion Planning Tool

Software Architecture & Implementation

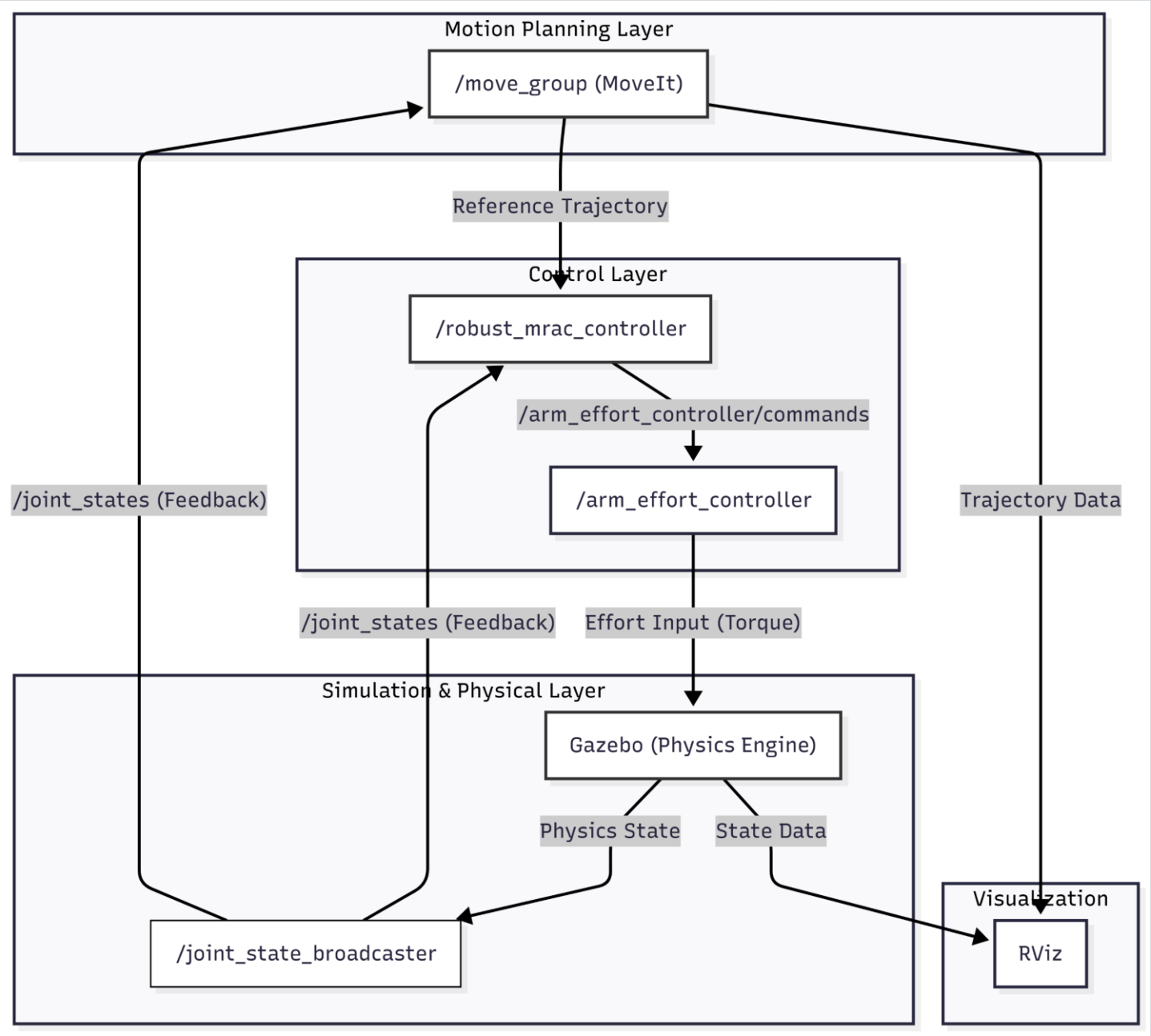
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- **Programming**: C++ (Controller), Python (Commander & Monitor).
- **Main Node**: `**robust_mrac_controller**` (1 kHz loop).

Component	ROS Native	Custom Built
Nodes	`move_group`, `robot_state_publisher`, `gazebo_ros`	`robust_mrac_controller`, `mrac_monitor`, `my_robot_commander`
Topics	`/joint_states`, `/tf`, `/follow_joint_trajectory`	`/mrac/K_hat`, `/mrac/W_hat`, `/mrac/lyapunov_V`, `/mrac/error`
Logic	OMPL Planning, KDL Kinematics	Robust Adaptive Control Law, Basis Functions

- **Communication**: Custom ROS messages for adaptation diagnostics.





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ROS2

GAZEBO

C++

RViz

MoveIt2

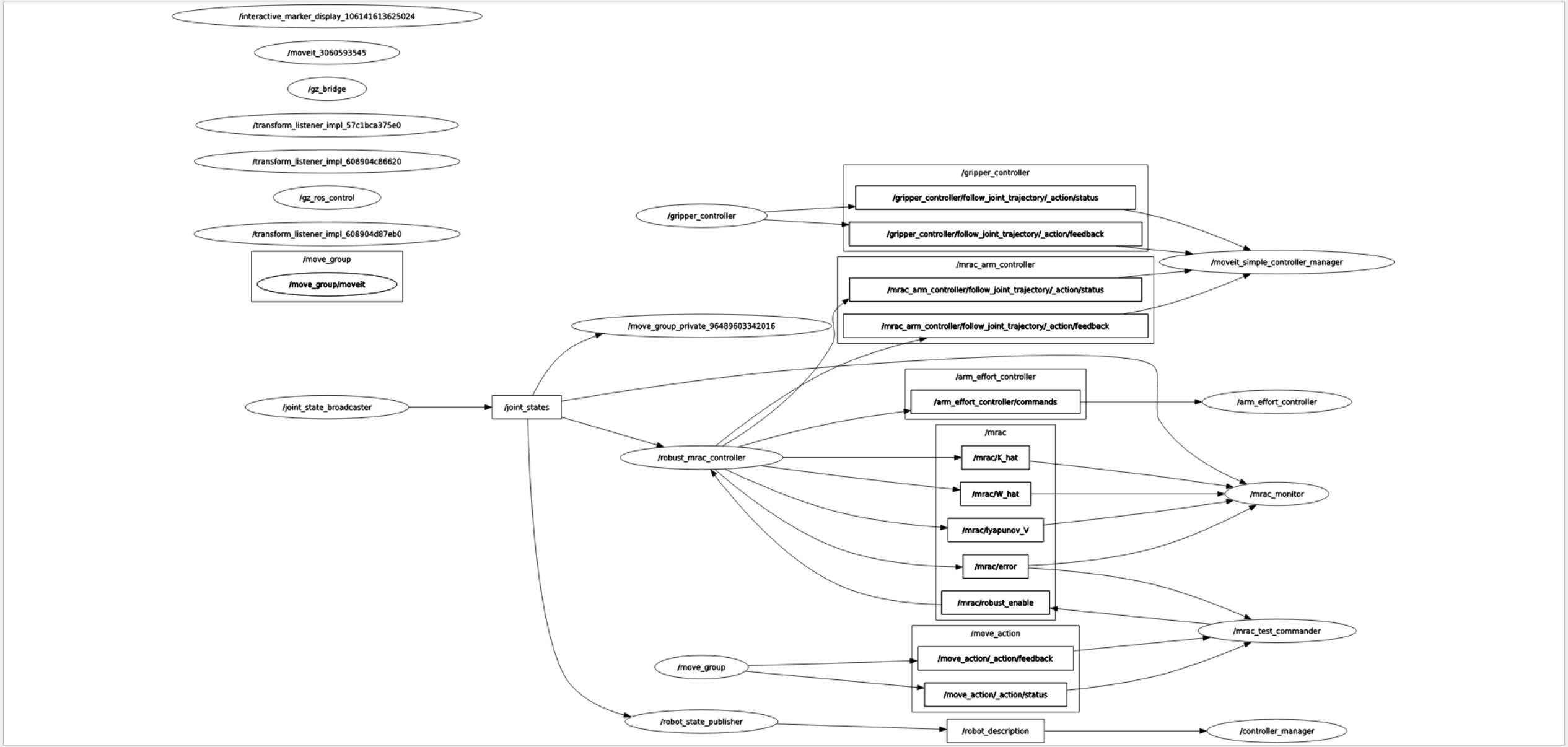
rqt_graph__RosGraph - rqt

Node Graph

Nodes/Topics (all) / /

Group: 1 Namespaces Actions tf Images Highlight Fit

Hide: Dead sinks Leaf topics Debug tf Unreachable Params



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Monitoring & Diagnostics

- **Custom Tool**: `mrac_monitor.py`.
- **Features**: Real-time matplotlib plots.
- **Tracks**: Lyapunov Energy (V), Tracking Error (e) and Adaptation Gains (ƙ).
- **Importance**: Verification of stability and convergence.





ROS2

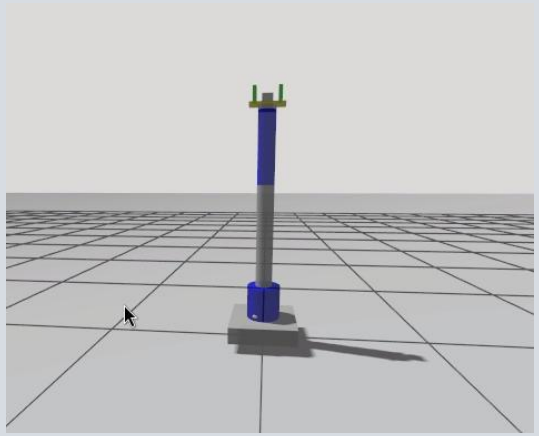
GAZEBO

RViz

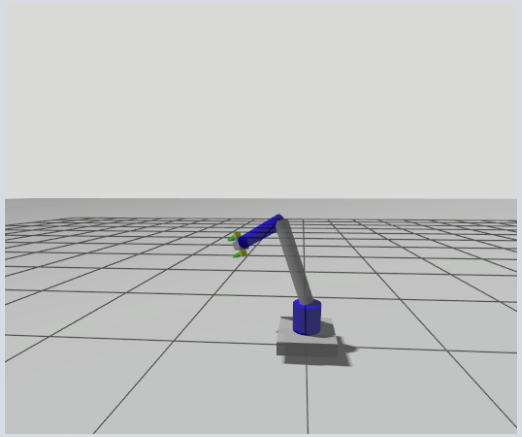
MoveIt2

Simulation Setup

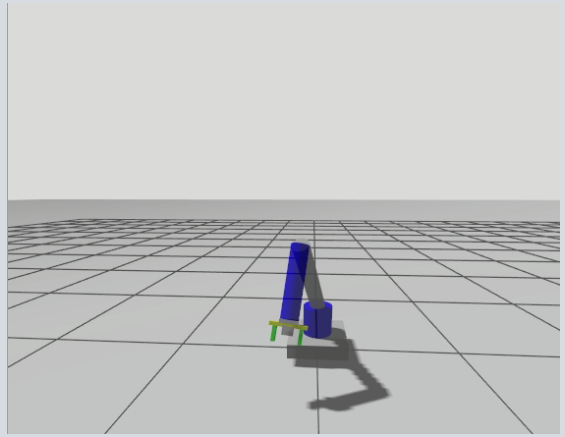
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Home



Pose 1



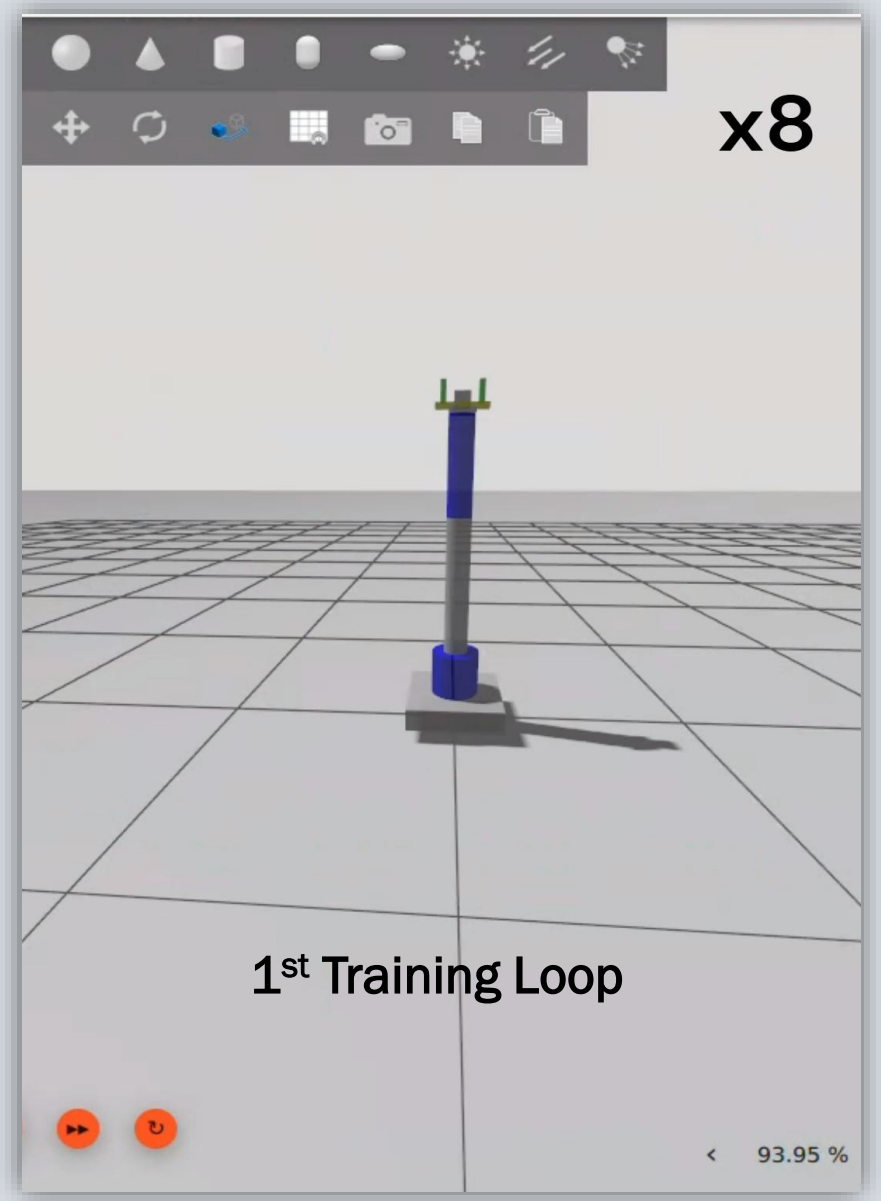
Pose 2

- The simulation ran for four training loops covering 3 positions configured using Rviz.
- The moveit is used for planning trajectory between each position.
- The poses were changed in a loop each of duration 170 seconds and pose switching at average 65 sec including 10 second hold.

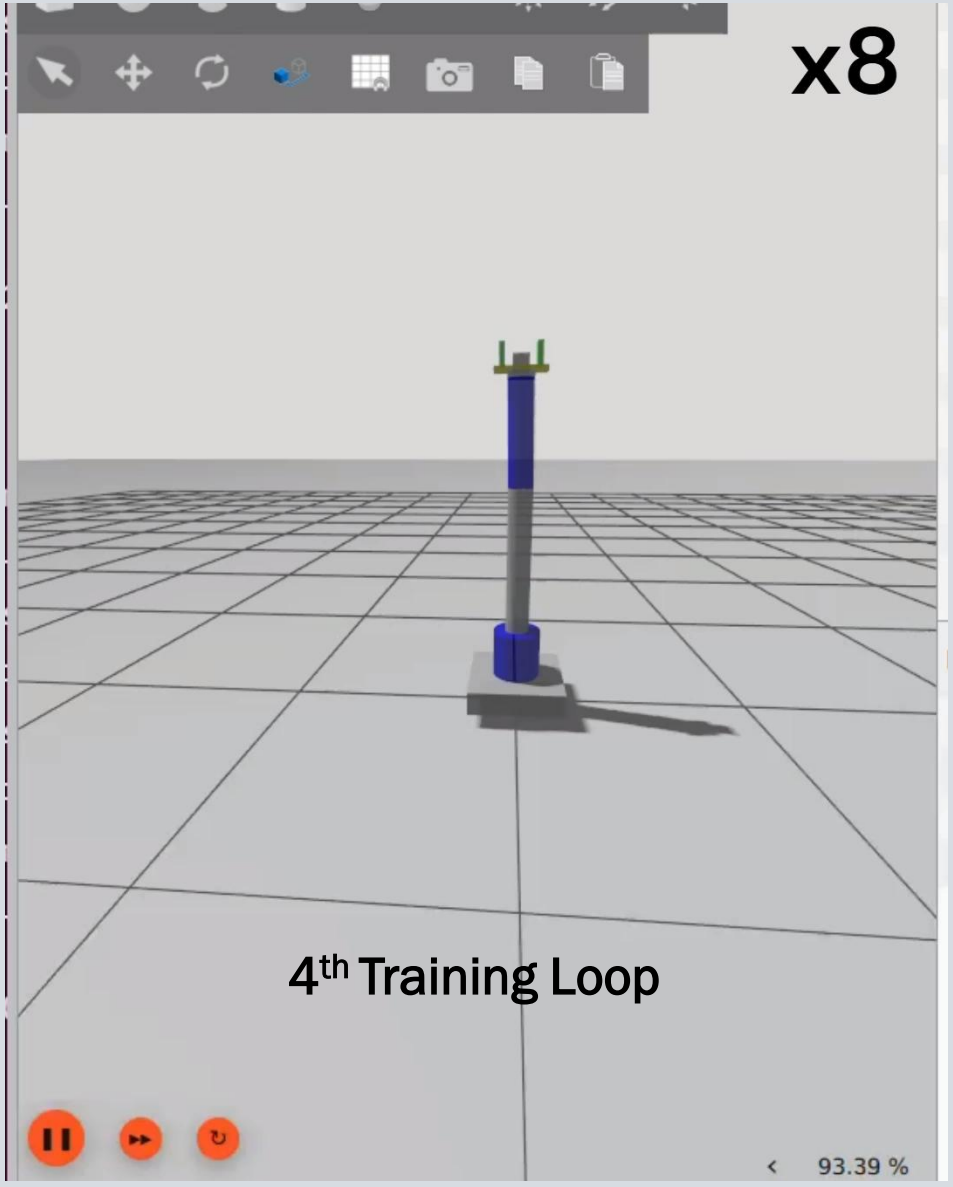
Poses	Joint 1	Joint 2	Joint 3	Joint 4
Home	0.0000	0.0000	0.0000	0.0000
Pose_1	-0.5031	0.3591	1.7818	0.2256
Pose_2	1.3358	0.7459	2.0133	1.2670

Simulation Preview

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1st Training Loop



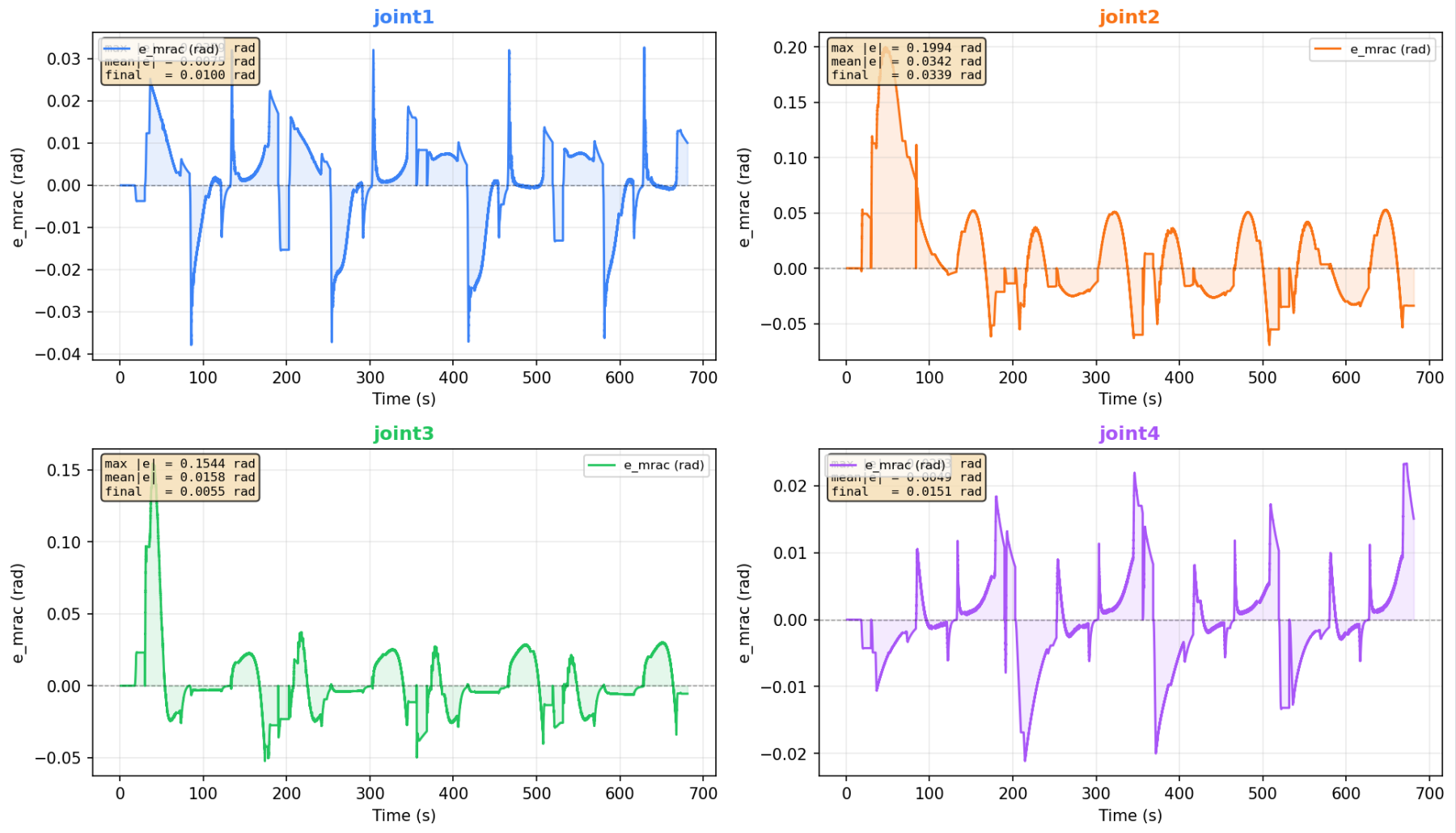
4th Training Loop

GAZEBO

Simulation Results

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Tracking error between reference model and plant -



Simulation Results

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Tracking error statistics for each joint

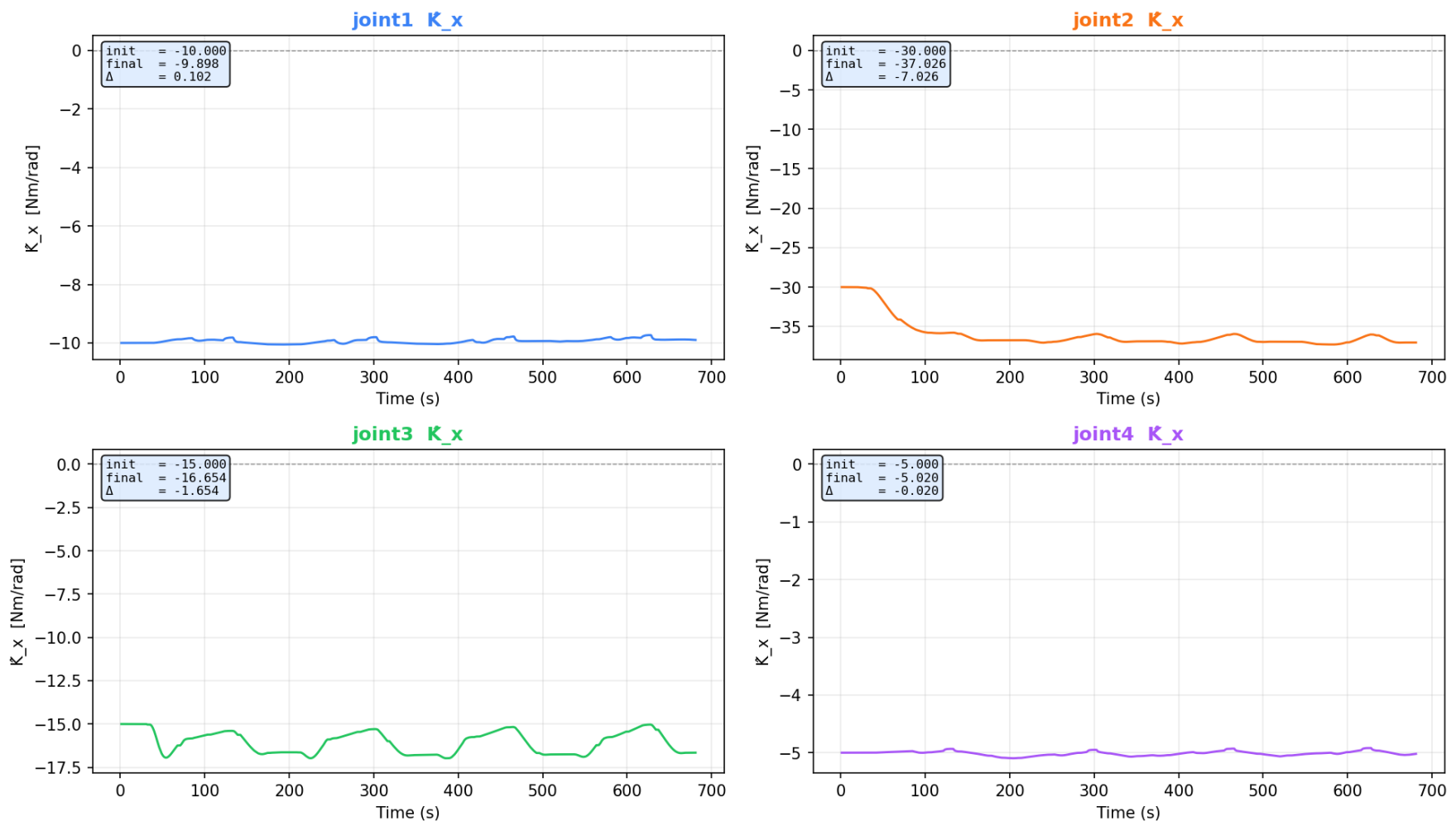
Metric	Joint 1	Joint 2	Joint 3	Joint 4
Max |e|(rad)	0.0322	0.1994	0.1544	0.0245
Mean |e| (rad)	0.0075	0.0342	0.0158	0.0049
Final e (rad)	0.0100	0.0339	0.0055	0.0151

- The error graphs clearly indicates that the robot was able to adapt to disturbances as error decreases with time.
- Joint 1 and 4 are rotational wrist joints i.e. are not moving against gravity, hence their errors are negligible.

Simulation Results

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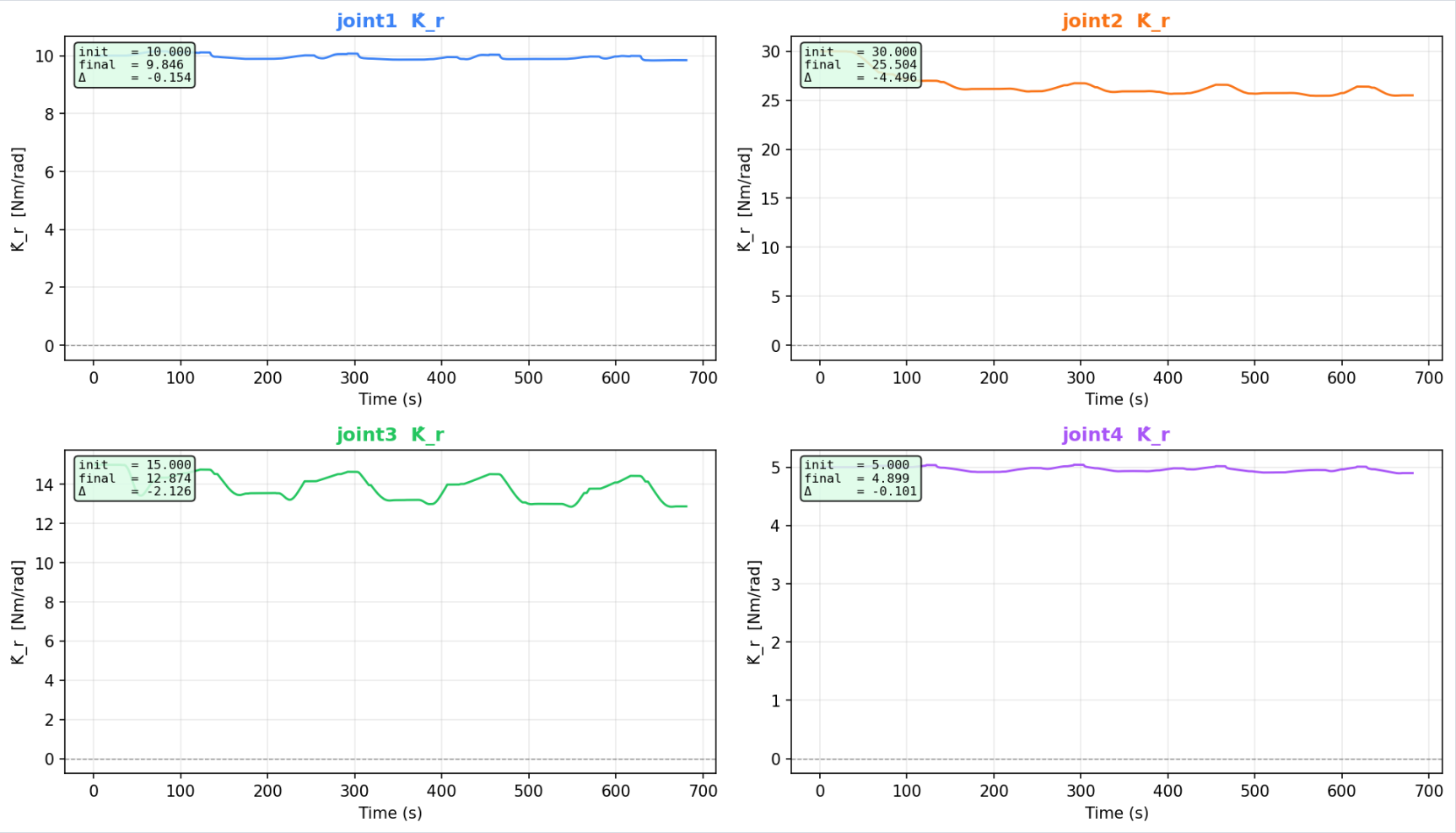
Adaptation of MRAC Gains -



Kx vs Time

Simulation Results

Adaptation of MRAC Gains -



Kr vs Time

- The gains adapts and settles in a narrow region.
- The larger oscillations in the joint 2 and 3 displays requirement of a robust controller for uncertainties.

Simulation Results

Adaptation of MRAC Gains -



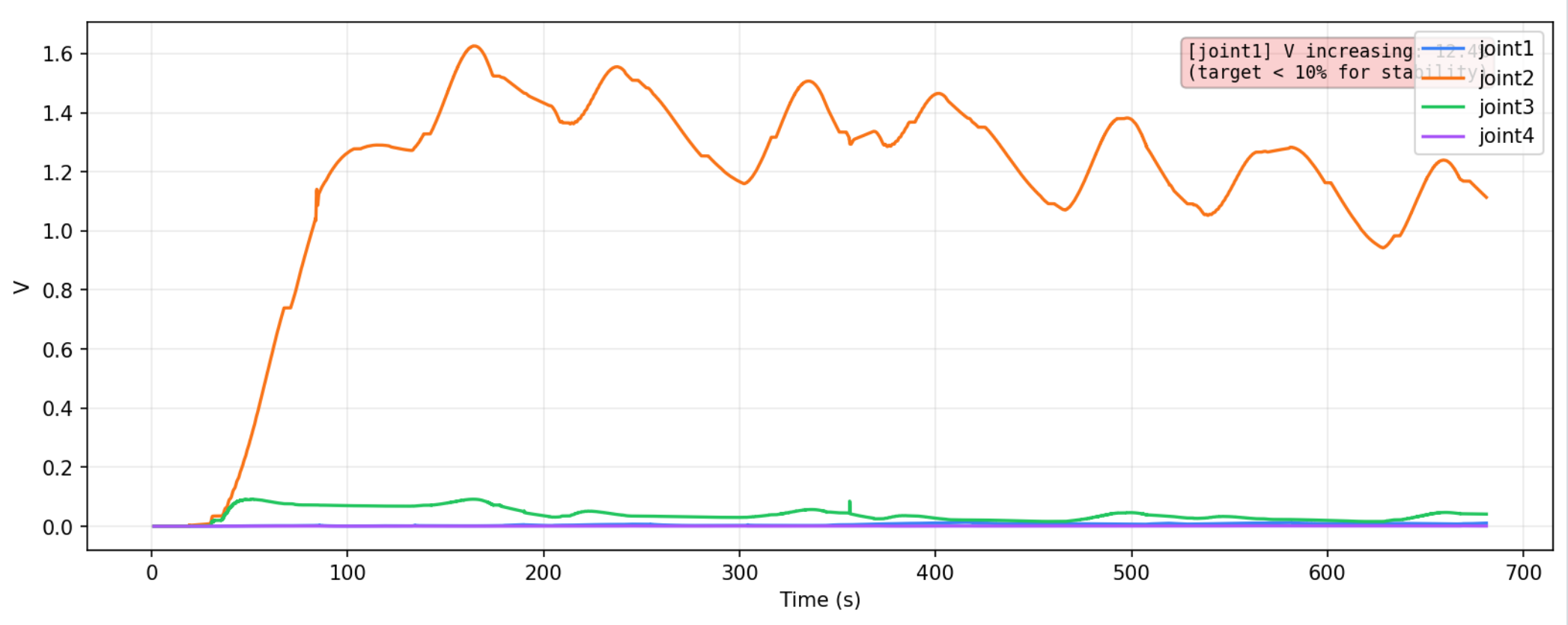
$|\hat{W}|$ vs Time

- The joints 1 and 4 gains are increasing, indicating the model is still adapting.

Simulation Results

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Adaptation of MRAC Gains -



- The joint 2 settles at a higher value indicating higher disturbance on base joint due to gravitational efforts.

Conclusion

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- The project successfully develop a 4-DOF robotic arm to track joint-space trajectories using an adaptive controller, without having to rely on a complex dynamical modeling of the robot.

Future Work

- The next step is to move from simulation to hardware i.e. implement the arm trajectory following using ros2control and moveit.
- Optional – We will try to implement the pick and place for different payloads.

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thank
you,